

Naraen Palanikumar

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PROFESSIONAL SUMMARY

Bioinformatics engineer with 4+ years of experience building scalable platforms for lab research. Proficient in designing Nextflow pipelines, Google Cloud architectures, and bioinformatics workflows. Proven track record of optimizing data analysis for peer-reviewed publications and deploying production-grade bioinformatics solutions on Google Cloud, while leveraging AI to automate complex data interpretation and accelerate scientific discovery.

EDUCATION

University of Illinois - Urbana-Champaign	Aug 2023 - May 2024
Master of Engineering (MEng), Biomedical Engineering	
University of Illinois - Urbana-Champaign	Aug 2019 - May 2023
Bachelor of Science (BS), Biomedical Engineering	

PROFESSIONAL EXPERIENCE

Phalaena Automata	Mar 2026 - Present
Founding Engineer Austin, TX, USA	
- Spearheading end-to-end development of a biotechnology research accelerator platform, managing the product lifecycle from initial architecture to cloud deployment. - Developing full-stack AI-centric infrastructure on Google Cloud, integrating RAG + LLM architecture to automate unstructured data interpretation pipelines and accelerate the sequence-to-insight loop.	
Independent	Jul 2025 - Feb 2026
Bioinformatics Consultant Remote	
- Consulted on single-cell sequencing pipelines, metabolic pathway analysis, and cellular heterogeneity queries to provide scalable workflows for research teams. - Applied cloud-native bioinformatics pipelines built for high-throughput omics research, leveraging a professional data engineer certification from Google Cloud.	
Malloy Lab @ Uniformed Services University of the Health Sciences	Oct 2024 - Jun 2025
Research Assistant Bethesda, MD, USA	
- Processed study participant biosamples and metadata for downstream high-throughput processing using flow cytometry and scRNA-seq. - Led bioinformatics analysis to extract and analyze CD4 T-cell repertoires from scTCR-seq data, contributing to published research in the journal <i>Frontiers in Immunology</i>.	
Sweeney Lab @ University of Illinois at Urbana-Champaign	Aug 2021 - Jun 2023
Undergraduate Research Assistant Champaign, IL, USA	
- Analyzed complex scRNAseq datasets using Cell Ranger and Seurat to uncover metabolism-associated neural circuits and rare cell populations. - Built and launched an interactive web application using R Shiny to streamline fiber photometry data processing and interpretation, contributing to published research in <i>The Journal of Neuroscience</i>.	

PROJECTS & OUTSIDE EXPERIENCE

Bench @ Phalaena Automata Python, FastAPI, React, Nextflow, Docker, GCP, Vertex AI	Mar 2026 - Present
- Building a production omics platform that ingests varied sequencing data into a centralized API and runs containerized Nextflow workflows on Cloud Run with strict IAM. Integrates a tool-augmented RAG loop over Vertex AI to cross-reference output results with PubMed for accelerated insights. Link to project	
AML scRNA-seq Analysis R, Python, R Shiny, Seurat, Scanpy	Sep 2025 - Oct 2025
- Built a reproducible scRNA-seq pipeline on 38 public AML patient samples, performing QC, anchor-based integration, and reference-guided cell-type annotation to characterize pre-leukemic populations. Extended with pseudotime, fate mapping, and survival analysis of PLPS/Stem11 signatures against NCI clinical data and deployed findings via an interactive R Shiny dashboard. Link to project	

SKILLS

Programming & Data Science: Python, R, SQL, MATLAB, JavaScript, TypeScript, HTML/CSS, React, Bash
Bioinformatics & Genomics: Bioinformatics, Genomics, Transcriptomics, Proteomics, Metabolomics, 10x Genomics, Nextflow
Cloud & Infrastructure: Google Cloud Platform, FastAPI, Data Engineering, Cloud-Native Architecture, Docker
AI & Machine Learning: Machine Learning, Pytorch, Clustering, UMAP, LLM/RAG Architecture, Vertex AI